Rushikesh R. Lagad - Curriculum Vitae

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Research Profile

I am a second-year PhD student in genomics and bioinformatics. My work asks which signals in microbial and microbiome data still hold up after careful testing, and which only appear to.

My first-author paper showed that the way genes are arranged inside genomic islands helps separate disease-causing *Enterococcus cecorum* from harmless strains in poultry. I now bring the same approach to gut microbiome studies in livestock and human disease: interpretable models, validation across independent cohorts, and causal inference.

Before Arkansas I worked in cancer bioinformatics at Moffitt Cancer Center. I trained first as a pharmacist, which keeps me focused on results that people outside the lab can use.

Education

PhD, Genomics and Bioinformatics, University of Arkansas, Fayetteville, AR. July 2025 to present (anticipated May 2028). Dale Bumpers College of Agricultural, Food and Life Sciences. Advisor: Dr. Aranyak Goswami. Focus: microbial comparative genomics, interpretable machine learning, and causal inference in host-microbiome systems.

MS, Bioinformatics and Computational Biology, University of South Florida, Tampa, FL. January 2022 to December 2024. Cancer genomics, single-cell and spatial transcriptomics, multi-omics integration.

MS, Pharmaceutical Nanotechnology, Taneja College of Pharmacy, University of South Florida, Tampa, FL. 2021 to 2022. Targeted drug delivery and metabolite-nanoparticle interaction modeling.

BS, Pharmacy, Savitribai Phule Pune University, Pune, India. June 2016 to May 2020. Pharmacology, molecular biology, clinical trials, and pharmacokinetics.

Research Experience

Graduate Research Assistant, Goswami Lab, Department of Animal Science and Center for Agricultural Data Analytics, University of Arkansas System Division of Agriculture. July 2025 to present

- Conceived and built Cassette2Vec-EC, a method that reads the arrangement of genes inside genomic islands to classify poultry-associated *Enterococcus cecorum*. On 145 genomes it reached an AUROC of 0.918 in genome-grouped cross-validation and 0.694 when whole studies were held out. Published in *Frontiers in Microbiology* (2026).
- Lead the computational work on a cross-cohort study of gut microbiome signatures in depression, using leave-one-study-out validation and Mendelian randomization to separate reproducible signal from cohort artifacts.

- Reanalyzed a public beef cattle rumen multi-omics dataset (PRJNA448333) to test whether reported feed-efficiency associations reproduce, and built a Mendelian randomization and variational autoencoder pipeline for it.
- Serve as computational lead for growth-performance analysis in an Arkansas Biosciences Institute study of gut microbiome and immune co-development in swine (126 pigs, three dietary regimens).
- Co-developed PDAC-EcoMap, a contamination-aware framework for grading intratumoral microbiome evidence in pancreatic cancer, accepted for an oral presentation at Moffitt Cancer Center.
- Work with collaborators at Argonne National Laboratory on Arctic soil microbiome data and at the University of Nebraska-Lincoln on single-cell annotation of taste cells.
- Drafted the initial manuscript and figures for a review of AI and multi-omics in neuronal gene expression, published in *Brain Informatics* (2026).

Bioinformatics Analyst, Moffitt Cancer Center, Tampa, FL. July 2024 to June 2025

- Rebuilt multi-omics workflows in Nextflow and Snakemake for genomic, transcriptomic, and LC-MS/MS metabolomic data, cutting runtime by about 40 percent.
- Supported tumor ecosystem and spatial transcriptomics analyses for cancer research teams.

Graduate Research Intern, University of South Florida, Tampa, FL. July 2023 to June 2024

- Analyzed single-cell and spatial transcriptomics data (Seurat, Harmony) to characterize tumor microenvironments.
- Integrated metabolomics with transcriptomics to link metabolite levels to gene expression.

Publications

Peer-reviewed articles

1. **Lagad RR**, Rafi S, Goswami A. Genomic-island cassette architecture provides interpretable signal for exploratory classification of poultry-associated *Enterococcus cecorum* lineages. *Frontiers in Microbiology*. 2026;17:1882753. doi:10.3389/fmicb.2026.1882753.
2. Goswami A, **Lagad RR**, Rafi S. Decoding neuronal gene expression: integrative insights from omics and AI. *Brain Informatics*. 2026;13(1):36. doi:10.1186/s40708-026-00323-z.

Preprint

1. **Lagad RR**, Rafi S, Goswami A. Genomic-island cassette architecture drives pathogenic *Enterococcus cecorum* lineages: Cassette2Vec-EC, a structural genomics and machine-learning framework. *bioRxiv*. 2026. doi:10.64898/2026.02.21.707173. Earlier version of article 1.

Submitted and under review

1. **Lagad RR**, Tsai TC, Goswami A. From WGS to actionable insight: making pathogen bioinformatics more interpretable for clinical and public-health audiences. Invited clinical review, *Infectious Disease Special Edition*. Submitted.

2. **Lagad RR**, Goswami A. From association to causal prioritization in rumen microbiome research: multi-omics reanalysis and translational frameworks for feed efficiency and methane mitigation in beef cattle. *Animal Microbiome*. Under review.

In preparation

1. **Lagad RR**, et al. Moving from taxonomy to function does not sharpen microbiome inference: paired multi-omics evidence from beef cattle rumen and Arctic tundra soil.

Presentations

Talks

1. **Lagad RR**, Modha N, Pathak D, Goswami A. PDAC-EcoMap: contamination-aware evidence gating for leakage-robust, interpretable machine learning on the intratumoral microbiome in pancreatic ductal adenocarcinoma. Selected abstract talk, Session V (AI in Biology), Dr. Robert Gillies Machine Learning Workshop in Cancer, Moffitt Cancer Center, Tampa, FL. October 15 to 16, 2026.

Posters

1. Modha N, **Lagad RR**, Goswami A, Tsai TC, Howe S, Watson A, Patel S. Modeling gut microbiome-immune co-development using multi-omics and machine learning in swine. Arkansas Biosciences Institute Annual Research Symposium, Fayetteville, AR. October 16, 2026.
2. **Lagad RR**, Goswami A. Poster presentation. ASM Microbe 2026, American Society for Microbiology, Washington, DC. June 4 to 7, 2026.
3. **Lagad RR**, Rafi S, Goswami A. Cassette2Vec-EC: genomic-island cassette architecture for interpretable prediction of pathogenic poultry-associated *Enterococcus cecorum*. AR-BIC 2026, Arkansas Bioinformatics Consortium, North Little Rock, AR. May 18 to 19, 2026.
4. **Lagad RR**, Goswami A. Cassette2Vec-EC: island-aware, explainable machine learning for early prediction of pathogenic *Enterococcus cecorum* from NGS genomes. Abstract 252P, Student Competition, International Poultry Scientific Forum, Atlanta, GA. January 26 to 27, 2026.

Other presentations

- ML for pathogenicity prediction in poultry microbiomes. AI in Agriculture Symposium, Fayetteville, AR. 2025.
- Cassette2Vec: AI-based AMR cassette prediction. AR-BIC 2025, Arkansas Bioinformatics Consortium. 2025.
- Oral and poster presentations at an Indo-German conference (2019) and Indo-African scientific meetings (2018), Pune, India.

Funded Research and Proposal Development

Funded projects

- Arkansas Research Alliance Impact Grant Program, grant GR023530 (PI: A. Goswami), with additional support from Cobb-Vantress Genetics LLC. Role: graduate researcher. Designed and ran the genomics and machine-learning analysis behind the *Frontiers in Microbiology* paper.

- Arkansas Biosciences Institute, "Modeling Gut Microbiome-Immune Co-Development Using Multi-Omics and Machine Learning in Swine" (PI: A. Goswami). Role: computational lead for growth-performance statistics.

Proposal development

- Contributed drafting, preliminary analysis, and submission coordination for Goswami Lab proposals to NIH (NIMH R21 and the NIH Director's New Innovator Award), DOE, USDA Animal Health and Disease Research, and the DoD CDMRP.

Honors and Recognition

- Omni Endowed International Student Scholarship, University of Arkansas. 2026.
- Selected participant and project co-lead, NIAID-BRC AI Codeathon 2.0, Argonne National Laboratory. September 2026.
- Invited at ASM Microbe 2026 to write a clinical review for *Infectious Disease Special Edition*.
- Poster selected for the Student Competition, International Poultry Scientific Forum. January 2026.
- Graduate Research Assistantship, Department of Animal Science, University of Arkansas. 2025 to present.

Media coverage

- "Arkansas researchers look at 'genetic neighborhoods' to find pathogenic bacteria." University of Arkansas System Division of Agriculture. August 11, 2026. The release credits the idea behind the study to me.
- Also carried by Feedstuffs and News-Medical on August 12, 2026, and by The Microbiologist.

Leadership, Mentoring and Team Science

- **Project co-lead, Hypothesis2Omics**, NIAID-BRC AI Codeathon 2.0, Argonne National Laboratory. September 16 to 18, 2026. Co-led an eight-person, multi-institution team building an agent workflow that turns a biological hypothesis into a reproducible analysis of public omics datasets.
- **Team member, BreedForward**, AI in Ag Hackathon, University of Arkansas. September 2026. Built a decision tool for choosing which maize inbred lines to advance, with code on GitHub. Also competed in the 2025 hackathon.
- **Mentoring.** Guide junior lab members through microbiome pipelines (QIIME 2, shotgun metagenomics), figure preparation, and conference abstracts. Informally advise a pharmacy master's student on a dissertation project.
- **Lab infrastructure.** Built an automated tool in Python that tracks funding opportunities for the lab.

Technical Skills

Area	Tools and methods
Programming and computing	Python, R, Bash; Linux, Slurm HPC; Git and GitHub

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Workflow engineering	Nextflow, Snakemake, conda and mamba environments
Microbial genomics	Prokka, PIRATE, IslandViewer 4, ABRicate, RGI/CARD, eggNOG-mapper
Microbiome and metagenomics	QIIME 2, DADA2, fastp, MetaPhlAn 4, HUMAnN
Transcriptomics	DESeq2, limma, Seurat, Harmony, pySCENIC; single-cell and spatial data
Machine learning	Random Forest, XGBoost, elastic net, variational autoencoders, SHAP; grouped and leave-one-study-out validation
Causal inference	Two-sample Mendelian randomization, sensitivity analysis
Metabolomics	LC-MS/MS preprocessing, pathway enrichment, integration with RNA-seq

Memberships and Training

- American Society for Microbiology (ASM)
- Sigma Xi, The Scientific Research Honor Society
- Metabolomics Association of North America (MANA), through January 2027
- Training: QIIME 2 (2024.10), advanced Linux and HPC, pharmacovigilance